Work Breakdown Structure (WBS): Dahdi Game

Project Name: Dahdi — Traditional Indian Mills Board Game
Institution: Academic / College Game Development Project
Author / Lead Developer: Soumya Ranjan Samal
Collection: Traditional / Indigenous Strategy Game Project

Level	WBS Code	Element Name (Development Steps)	Definition (Clearly define work to be done)
1	1.0	**Project Startup & Framework Initialization**	Initial project scoping, technical requirements, source structure and browser development setup.
2	1.1	Game Concept & Rules Formalization	Define Dahdi objectives, nine-piece setup, 24 board points, mills, captures, movement, flying and win conditions.
2	1.2	Architecture Design & Module Separation	Separate application control, rules, AI, rendering, audio, constants and styles.
2	1.3	Source Organization & Development Setup	Maintain JavaScript modules and CSS files in organized project directories and test the game in a modern browser.
1	2.0	**Board & Core Game Mechanics**	Implement the playable 24-point board and the complete match state.
2	2.1	24-Point Board Configuration	Define the three concentric-square board positions and point coordinates.
2	2.2	Player Piece State	Create nine pieces per side and track pieces in hand and on the board.
2	2.3	Placement Phase	Allow players to place one available piece on an empty point during Phase 1.
2	2.4	Movement Phase	Generate and apply legal adjacent moves during Phase 2.
2	2.5	Flying Phase	Allow a player with exactly three pieces to move to any empty point during Phase 3.
2	2.6	Mill & Capture Rules	Detect three-in-a-row mills, identify removable opponent pieces and protect pieces in mills when required.
2	2.7	Win & Draw Conditions	Detect opponent reduction to two pieces and situations with no legal movement.

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1	3.0	**Artificial Intelligence (AI) Opponent**	Develop the local computer opponent and difficulty settings.
2	3.1	Easy AI	Use shallow search and some random choices for a casual challenge.
2	3.2	Medium AI	Use depth-3 search and move ordering for a balanced challenge.
2	3.3	Master AI	Use deeper search, iterative/deeper analysis and transposition caching.
2	3.4	Position Evaluation	Evaluate material, mills, threats and mobility.
2	3.5	Hint System	Use the AI decision system to provide a useful player move hint.
1	4.0	**Art, Visuals & Game Feedback**	Create the visual board, tokens, indicators and celebratory effects.
2	4.1	Board Rendering	Render the 24-point board and selectable positions.
2	4.2	Player Tokens	Display Peacock and Haldi pieces and player status.
2	4.3	Legal Move Indicators	Highlight selectable pieces and valid destination points.
2	4.4	Victory Feedback	Trigger win confetti and result feedback.
1	5.0	**Audio & Settings**	Implement game audio, music and player preferences.

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2	5.1	Web Audio Engine	Initialize audio context and separate SFX/music gain controls.
2	5.2	Sound Effects	Provide UI and gameplay feedback sounds.
2	5.3	Tanpura Music	Provide optional ambient Indian Tanpura drone.
2	5.4	Preference Storage	Store mute, volume, language, colour-blind and reduced-motion preferences in local storage.
1	6.0	**User Interface & User Experience**	Build menus, settings, help, HUD and accessibility presentation.
2	6.1	Game Mode Selection	Provide Player vs AI and 2-Player Pass & Play.
2	6.2	AI Difficulty Selection	Provide Easy, Medium and Master.
2	6.3	Board Variant	Provide the optional diagonal configuration.
2	6.4	Language Selection	Provide English, Hindi and Odia.
2	6.5	Rules / Help Screen	Explain Placement, Movement and Flying.
2	6.6	Undo / Hint / Rematch	Provide utility controls for player assistance and replay.
1	7.0	**Quality Assurance & Final Submission**	Verify gameplay and prepare final academic documentation.
2	7.1	Rules Testing	Test placement, movement, flying, mills, captures and win conditions.

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2	7.2	AI Testing	Test all AI levels and ensure only legal moves are selected.
2	7.3	UI Testing	Test settings, language, audio, accessibility, undo, hint and result states.
2	7.4	Browser Testing	Check rendering and interaction in modern browsers.
2	7.5	Documentation	Prepare GDD, Pitch Proposal, Project Charter, WBS, Journal and Modification Task Sheet.